

LESSON 7-7 • SESSION 1 • PRACTICE • FORM A

Polygons and Perimeter on the Plane

Learning target – I can find the perimeter and area of a polygon drawn on the coordinate plane. · 6.G.A.3

REMEMBER

- A **polygon** is a closed shape with straight sides. Its corners are **vertices**.
- Find each side from matching coordinates: opposite signs **add**, same sign **subtract**.
- **Perimeter** = add all the sides. **Area** of a rectangle = length $\times$ width.

WORKED EXAMPLE

A rectangle has vertices $(-2, 3)$, $(5, 3)$, $(5, -2)$ and $(-2, -2)$. Each unit is 1 meter. Find its perimeter.

Step 1 Top side: same y , so $2 + 5 = 7$ m.

Step 2 Right side: same x , so $3 + 2 = 5$ m.

Step 3 Add all four sides: $7 + 5 + 7 + 5 = 24$.

Answer Perimeter = 24 m

WORDS polygon (*polígono*) · vertex (*vértice*) · perimeter (*perímetro*)

PRACTICE

show your work · label units

1. A rectangle has vertices $(-3, 4)$, $(6, 4)$, $(6, -2)$ and $(-3, -2)$. Find each measure.

a. side $(-3, 4)$ to $(6, 4)$ _____

b. side $(6, 4)$ to $(6, -2)$ _____

c. perimeter _____

d. area _____

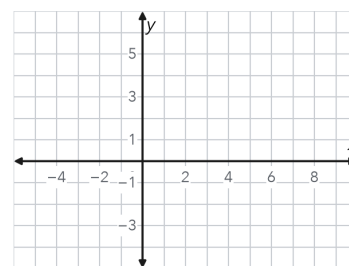
2. **Garden Plot.** A garden plot has corners at $(-4, 6)$, $(9, 6)$, $(9, -3)$ and $(-4, -3)$. Each unit is 1 meter.

a. Plot the corners and connect them.

b. Top side _____ m

c. Left side _____ m

d. Perimeter _____ m



3. **Dog Run.** Dog run vertices: $(-7, 10)$, $(11, 10)$, $(11, -1)$ and $(-7, -1)$. Each unit is 1 meter. Find the perimeter.

Hint Find one side across and one side up and down.

Show your work.

Answer _____ m

4. **Missing Vertex.** Three corners of a rectangle are $(-2, 9)$, $(8, 9)$ and $(8, 1)$. Name the fourth corner.

Hint Match x with one corner and y with another.

Show your work.

Corner _____

5. **Right Triangle.** A right triangle has vertices $(-5, 2)$, $(7, 2)$ and $(7, -5)$. Find its area.

Hint Area = $1/2 \times \text{base} \times \text{height}$.

Show your work.

Answer _____ sq. units

6. **Deck Area.** Deck vertices: $(-6, -2)$, $(7, -2)$, $(7, -9)$ and $(-6, -9)$. Each unit is 1 yard. Find its area.

Hint Both y -values are negative, so subtract to find the height.

Show your work.

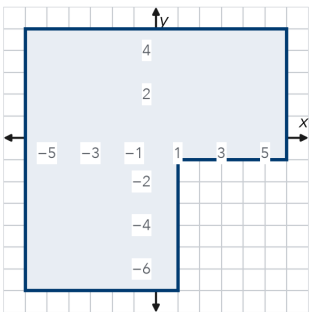
Answer _____ sq. yd

7. **Mural Outline.** The mural outline has vertices $(-6, 5)$, $(6, 5)$, $(6, -1)$, $(1, -1)$, $(1, -7)$ and $(-6, -7)$. Each unit is 1 foot. Find the perimeter.

Hint Go around in order. Each side joins two vertices that share a coordinate.

Show your work.

Answer _____ ft



8. **Fence and Sod.** A plot has vertices $(-6, 5)$, $(9, 5)$, $(9, -4)$ and $(-6, -4)$. Fencing costs \$7 per unit. Sod costs \$2 per square unit.

- a. Find the length and the width. _____
- b. Find the perimeter and the cost of fencing. _____
- c. Find the area and the cost of sod. _____

FIND THE ERROR explain the mistake, then fix it

9. **Jorge's Work.** For a side from $(-5, 4)$ to $(3, 4)$, Jorge wrote the length as $5 - 3 = 2$ units. Explain his mistake. Then find the correct length.

TEST PRACTICE choose the best answer

10. A rectangle with sides 8 units and 5 units has a perimeter of –
A 13 units **B** 26 units **C** 40 units **D** 80 units
11. A rectangle with sides 8 units and 5 units has an area of –
A 13 square units **B** 26 square units **C** 40 square units **D** 80 square units

SELF-CHECK CODE find your answer · copy its letter into that problem's box · spell a math word

143 J	\$168 K	29 Q	$(-2, 1)$ E	58 V	42 R	48 E	\$336 X	91 T	84 Z
#3		#4		#5		#6		#7	
								#8b	

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____